



Build a Virtual Private Cloud (VPC)



Damien Joseph

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional [Info](#)
Creates a tag with a key of 'Name' and a value that you specify.

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy [Info](#)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional
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Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual network within AWS that lets you launch resources securely. It's useful because it provides full control over networking, including IP addressing, routing, and connectivity.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create a secure, isolated network environment, manage subnets, control traffic routing, and enable internet access for instances through an Internet Gateway.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was how crucial proper subnet configuration and routing were. A small mistake in settings could completely block internet access to my instances.



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This project took me...

This project took me 45 minutes.



Virtual Private Clouds (VPCs)

VPCs are isolated sections of the Aws Cloud that helps to keep my Aws resources private and secure.

There was already a default VPC in my account ever since my AWS account was created. This is because Aws has set up a default VPC to allow me to deploy resources like EC2 instances/RDS databases right away.

To set up my VPC, I had to define an IPv4 CIDR block, which means a range of IP addresses that my VPC can allocate to the resources deployed into my VPC,

The screenshot shows the 'Create VPC' page in the AWS Management Console. The page is titled 'Create VPC' and includes a sub-header 'Resources to create'. Under 'Resources to create', there are two radio buttons: 'VPC only' (selected) and 'VPC and more'. Below this, there is a section for 'Name tag - optional' with a text input field containing 'NextWork VPC'. The 'IPv4 CIDR block' section has two radio buttons: 'IPv4 CIDR manual input' (selected) and 'IPAM-allocated IPv4 CIDR block'. Below this, there is a text input field for 'IPv4 CIDR' containing '10.0.0.0/16'. The 'IPv6 CIDR block' section has three radio buttons: 'No IPv6 CIDR block' (selected), 'IPAM-allocated IPv6 CIDR block', and 'Amazon-provided IPv6 CIDR block'. Below this, there is a dropdown menu for 'Tenancy' set to 'Default'. At the bottom, there is a 'Tags' section with a table for adding tags.

Key	Value - optional
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Subnets

Subnets are subsections of my VPC, just like how neighbourhoods are subsections of a city. There are already subnets existing in my account, one for every AZ in the region that i've set up my VPC in.

After creating my subnet, I enabled auto-assign public IPv4 to ensure instances automatically get public IPs, allowing direct internet access and simplifying deployment for public-facing resources like web servers.

The difference between public and private subnets are Public subnet have a route to Internet Gateway and Private subnet doesn't have a route to Internet gateway. For a subnet to be considered public, it has to have a route to an Internet gateway.



You have successfully changed subnet settings.
Enable auto-assign public IPv4 address.

Subnets (1/1) info

Find subnets by attribute or tag

Subnet ID: subnet-039a7acf668bd733 X Clear filters

Last updated less than a minute ago Actions Create subnet

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR	IPv6 CIDR association ID	Avail
Public 1	subnet-039a7acf668bd733	Available	vpc-0e141b52a56c9287a Nex...	Off	10.0.0.0/24	-	-	251



Internet gateways

Internet gateways are the key VPC components that allows internet access for the resources in my VPC/subnet. An IGW is also how users in the public can access my resources in the public subnet.

Attaching an Internet Gateway to a VPC enables resources in the public subnet to access the internet. Without it, users can't reach any instances or services hosted in that subnet.

	Name	Internet gateway ID	State	VPC ID	Owner
☐	-	igw-04b5d4a8692f922af	Attached	ypc-09080115c113f5a4	128200128581
☐	NextWork IG	igw-088d5574b6ec0e3fa	Attached	ypc-0d141b52a56c9287a NextWork VPC	128200128581



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